

Enterprise Multi-Site LAN Integration with Redundancy and Internet Connectivity

1. Project Title

Enterprise Multi-Site LAN Integration with:

- eBGP Routing
- HSRP Redundancy
- NAT/PAT Internet Access
- Dual CORE Architecture
- WAN Failover and High Availability

2. Project Objective

The objective of this project is to design and implement an enterprise-grade WAN infrastructure connecting multiple geographically separated office LANs.

The network was designed to provide:

- Inter-office communication
- Redundant routing paths
- Automatic failover
- Gateway redundancy
- Secure Internet access
- Dynamic route advertisement
- High availability and scalability

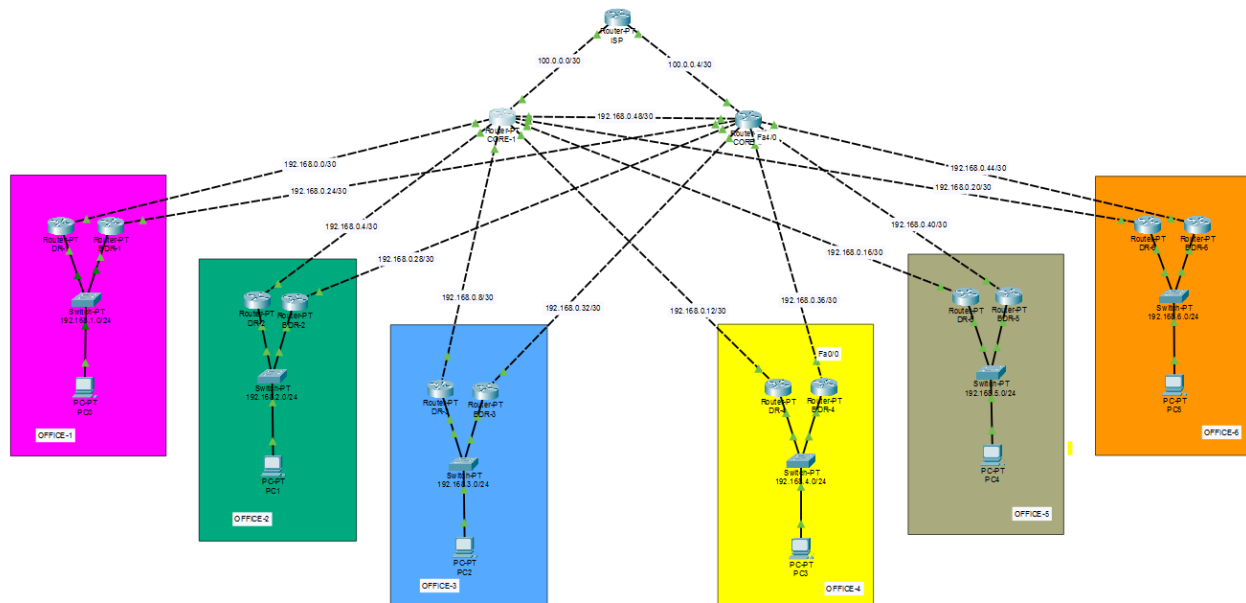
3. Technologies Used

Technology	Purpose
Cisco Packet Tracer	Network Simulation
eBGP	Dynamic Routing
HSRP	Gateway Redundancy
NAT/PAT	Internet Access
Static Routing	Default Routing
IPv4 Addressing	Network Communication

4. Network Components

Device Type	Quantity
ISP Router	1
CORE Routers	2
Distribution Routers (DR)	6
Backup Distribution Routers (BDR)	6
Switches	6
PCs	6

5. Network Topology Description



https://drive.google.com/file/d/1Dhw9hZJv_olVjs7wBj4-zHpdqZssjqSE/view?usp=sharing

The topology consists of:

- One ISP router providing public WAN connectivity.
- Two CORE routers connected to the ISP.
- Six branch offices.
- Each branch contains:
 - One DR router
 - One BDR router
 - One switch
 - One PC

The DR routers connect to CORE-1. The BDR routers connect to CORE-2.

HSRP was implemented between DR and BDR routers to provide gateway redundancy.

BGP was configured between:

- ISP ↔ CORE routers
- CORE routers ↔ Branch routers
- CORE-1 ↔ CORE-2

NAT/PAT was configured on CORE routers to allow all internal private networks to access the Internet.

6. IP Addressing Scheme

WAN Networks

Connection	Network
ISP ↔ CORE-1	100.0.0.0/30
ISP ↔ CORE-2	100.0.0.4/30
CORE-1 ↔ DR-1	192.168.0.0/30
CORE-1 ↔ DR-2	192.168.0.4/30
CORE-1 ↔ DR-3	192.168.0.8/30
CORE-1 ↔ DR-4	192.168.0.12/30
CORE-1 ↔ DR-5	192.168.0.16/30
CORE-1 ↔ DR-6	192.168.0.20/30
CORE-2 ↔ BDR-1	192.168.0.24/30
CORE-2 ↔ BDR-2	192.168.0.28/30
CORE-2 ↔ BDR-3	192.168.0.32/30
CORE-2 ↔ BDR-4	192.168.0.36/30
CORE-2 ↔ BDR-5	192.168.0.40/30
CORE-2 ↔ BDR-6	192.168.0.44/30
CORE-1 ↔ CORE-2	192.168.0.48/30

LAN Networks

Office	LAN Network	Virtual Gateway
OFFICE-1	192.168.1.0/24	192.168.1.1
OFFICE-2	192.168.2.0/24	192.168.2.1
OFFICE-3	192.168.3.0/24	192.168.3.1
OFFICE-4	192.168.4.0/24	192.168.4.1
OFFICE-5	192.168.5.0/24	192.168.5.1
OFFICE-6	192.168.6.0/24	192.168.6.1

7. Autonomous System (AS) Allocation

Device	AS Number
ISP	65000
CORE-1	65001
CORE-2	65002
DR-1	65011
DR-2	65012
DR-3	65013
DR-4	65014
DR-5	65015
DR-6	65016
BDR-1	65017
BDR-2	65018
BDR-3	65019
BDR-4	65020
BDR-5	65021
BDR-6	65022

8. ISP Router Configuration Summary

The ISP router was configured with:

- Public WAN interfaces
- eBGP neighbors toward CORE routers

Configured Features:

- eBGP Peering
- WAN connectivity
- Public IP communication

9. CORE Router Configuration

CORE-1 Functions

CORE-1 performs:

- WAN connectivity to ISP
- NAT/PAT services
- eBGP routing

- Connectivity toward DR routers
- Internal route exchange

Configured Features:

- NAT overload
- BGP neighbors
- Default routing
- WAN interface addressing
- Internet access

CORE-2 Functions

CORE-2 performs:

- Backup WAN connectivity
- NAT/PAT redundancy
- eBGP routing
- Connectivity toward BDR routers
- Route failover support

Configured Features:

- NAT overload
- BGP neighbors
- WAN redundancy
- Default route
- Backup Internet access

10. Distribution Router (DR) Configuration

Each DR router was configured with:

- WAN interface toward CORE-1
- LAN interface toward office switch
- HSRP active configuration
- eBGP neighbor relationship
- Default route toward CORE-1

Configured Features:

- HSRP Active Gateway
- eBGP LAN advertisement
- WAN communication
- Redundant routing

Example:

Router	LAN Gateway
DR-1	192.168.1.1
DR-2	192.168.2.1
DR-3	192.168.3.1
DR-4	192.168.4.1
DR-5	192.168.5.1
DR-6	192.168.6.1

11. Backup Distribution Router (BDR) Configuration

Each BDR router was configured with:

- WAN interface toward CORE-2
- LAN interface toward office switch
- HSRP standby configuration
- eBGP neighbor relationship
- Default route toward CORE-2

Configured Features:

- HSRP Standby Gateway
- Backup WAN connectivity
- eBGP route advertisement
- Automatic failover

12. HSRP Implementation

HSRP was implemented between DR and BDR routers in each branch office.

Purpose of HSRP:

- Gateway redundancy
- Automatic failover
- Continuous LAN connectivity

Configuration Logic:

- DR routers configured with higher priority (110)
- BDR routers configured with lower priority (100)
- Virtual IP used as default gateway for PCs
- Preemption enabled for automatic role recovery

Example:

```
standby 1 ip 192.168.1.1
standby 1 priority 110
standby 1 preempt
```

13. BGP Routing Implementation

External Border Gateway Protocol (eBGP) was implemented throughout the enterprise network.

Purpose:

- Dynamic route advertisement
- Redundant path selection
- WAN scalability
- Automatic rerouting

BGP Neighbor Relationships:

- ISP ↔ CORE Routers
- CORE Routers ↔ DR Routers
- CORE Routers ↔ BDR Routers
- CORE-1 ↔ CORE-2

Each DR/BDR router advertises its local LAN network.

Example:

```
router bgp 65011
neighbor 192.168.0.1 remote-as 65001
network 192.168.1.0 mask 255.255.255.0
```

14. NAT/PAT Implementation

NAT overload was configured on both CORE routers.

Purpose:

- Internet access for private networks
- Public IP conservation
- Secure internal addressing

Configuration Features:

- Inside interfaces configured on branch links
- Outside interface configured toward ISP
- Access-list permitting internal networks
- PAT overload enabled

Example:

```
access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface FastEthernet0/0 overload
```

15. Redundancy Features Implemented

The following redundancy mechanisms were implemented:

Feature	Purpose
Dual CORE Routers	WAN Redundancy
HSRP	Gateway Redundancy
BGP	Dynamic Failover
NAT Redundancy	Internet Continuity
Multi-WAN Paths	Fault Tolerance

16. Verification Commands

Verify Interfaces

```
show ip interface brief
```

Verify BGP

```
show ip bgp summary
show ip route bgp
show ip bgp
```

Verify HSRP

```
show standby brief
show standby
```

Verify NAT

```
show ip nat translations
show ip nat statistics
```

Connectivity Test

```
ping
```


traceroute

17. Failover Testing

The following scenarios were tested:

CORE Router Failure

Result:

- Traffic rerouted through backup CORE router.
- Internet connectivity remained active.

HSRP Failover

Result:

- BDR router became active automatically.
- Gateway continuity maintained.

WAN Link Failure

Result:

- BGP selected alternate available path.
- Communication remained uninterrupted.

18. Testing Results

The implemented network successfully achieved:

- Full inter-site communication
- Successful Internet access
- Dynamic route learning
- Automatic failover
- Gateway redundancy
- NAT translation
- BGP convergence

19. Advantages of the Design

Advantages:

- High availability

- Fault tolerance
- Scalable enterprise architecture
- Efficient routing
- Internet redundancy
- Automatic recovery mechanisms
- Secure private addressing

20. Conclusion

This project successfully implemented a complete enterprise multi-site WAN infrastructure using Cisco technologies.

Implemented Technologies:

- eBGP
- HSRP
- NAT/PAT
- WAN Redundancy
- Dual CORE Architecture

The network design ensures:

- Reliable communication
- Redundant Internet connectivity
- Automatic failover
- High availability
- Enterprise scalability

The project demonstrates practical implementation of enterprise-level routing, redundancy, and Internet integration using Cisco Packet Tracer.